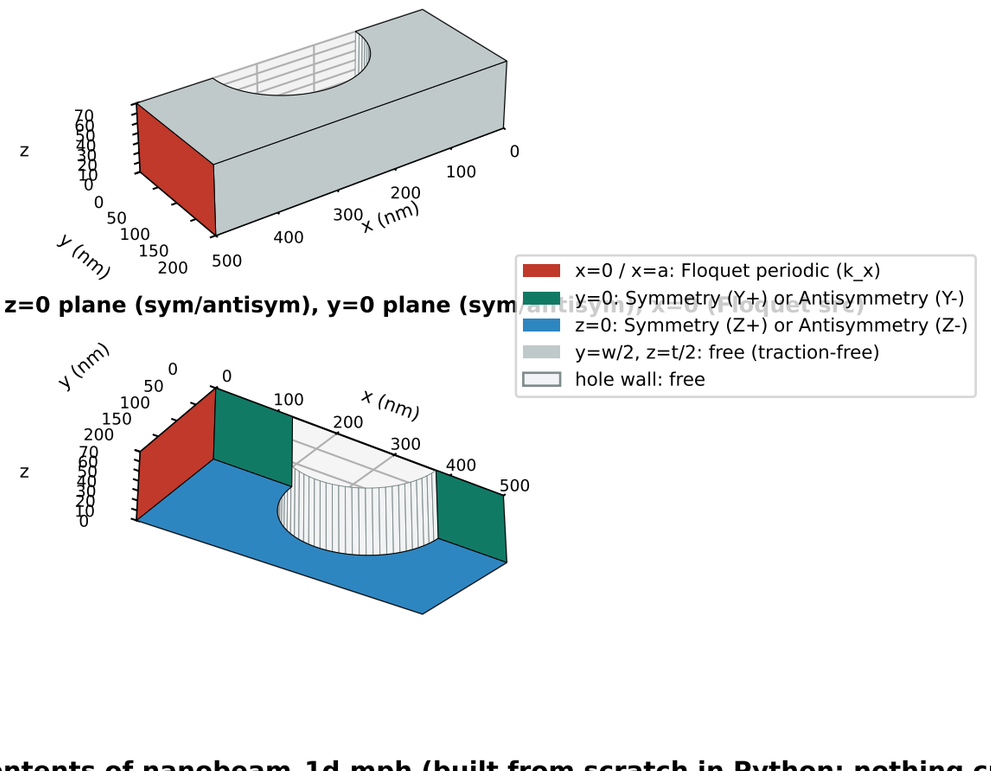
1D diamond nanobeam -- planned COMSOL model, for review before building

confirmed: w = 400 nm, t = 140 nm, air pad 1.5 um, quarter cell, 4 classes, mech on beam only | PLACEHOLDER a = 500 nm, hx/a = 0.5, hy/w = 0.6 (drawing only)

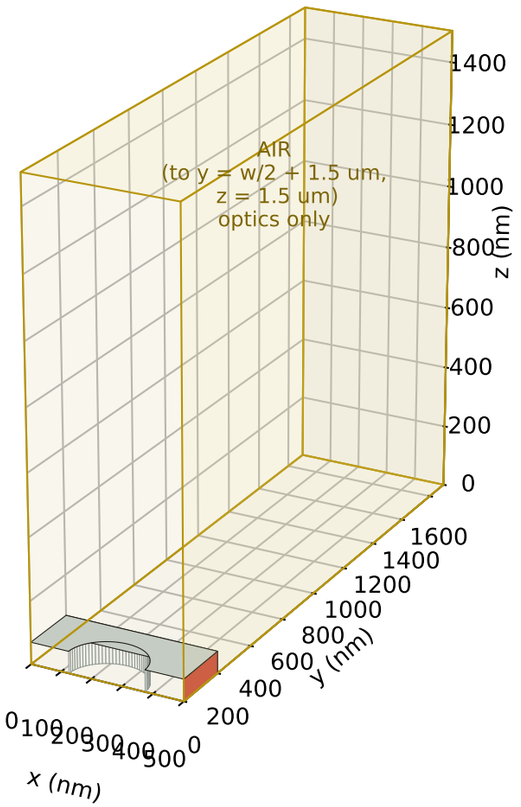
Beam quarter cell 0<x<a, 0<y<w/2, 0<z<t/2 (placeholder a, hx, hy; z drawn 1.6x stretched)

(A1) from ABOVE: top z=t/2 (free), x=a (Floquet dst), y=w/2 (free), hole wall



(A2) from BELOW: z=0 plane (sym/antisym), y=0 plane (sym/antisym), x=0 (Floquet src)

(B) Same quarter cell inside its air box, TO SCALE
(beam = small colored block at the corner; air has no mechanics)



(C) Planned contents of nanobeam_1d.mph (built from scratch in Python; nothing created yet)

Parameters	a [TBD], w = 400 nm, t = 140 nm, hx = (hx/a)*a, hy = (hy/w)*w [ratios TBD], pad_y = pad_z = 1.5 um, kx (Floquet), f_mech (shift), N_mode
Geometry	1) Block "beam": [0,a] x [0,w/2] x [0,t/2] 2) Work plane z=0: Ellipse center (a/2, 0), semi-axes hx/2, hy/2 -> Extrude to t/2 = "hole" 3) Difference beam - hole 4) Block "air": [0,a] x [0,w/2+pad_y] x [0,pad_z] 5) Form Union -> 2 domains (beam, air)
Materials	Diamond (isotropic): E = 1050 GPa, nu = 0.07, rho = 3500 kg/m^3, n = 2.41 -> beam domain; Air (n = 1) -> air domain
Solid Mechanics	domain = beam ONLY. Linear Elastic (isotropic) | Free (default, all other beam faces) | Periodic Condition: Floquet, k = (kx, 0, 0), source x=0 -> destination x=a | Symmetry + Antisymmetry on y=0 (one active per class) | Symmetry + Antisymmetry on z=0 (one active per class)
Meshes	Mech Mesh (beam only): Free Triangular on x=0 face -> Copy Face to x=a -> Free Tetrahedral; size PROPOSED hmax = t/5 (validated in 2D at a=1000) OP Mesh (beam + air): same periodic-face approach; sizes TBD with the optical study
Study "Mech"	Eigenfrequency (ARPACK, around shift f_mech), N_mode per class; Python loops kx = 0 .. pi/a (Gamma -> X) and the 4 classes (Y+Z+, Y-Z+, Y+Z-, Y-Z-) by activating the Symmetry / Antisymmetry features; extracts solid.freq
Not included yet	optical (ewfd) physics and study, outer-air optical BC, PML / Q calculations, cavity (tapered) geometry
Validation	first model run: hole disabled (hx = hy = 0) -> compare lowest bands with analytic beam dispersion (longitudinal sqrt(E/rho), flexural, torsion)